

The background image shows a man in a light blue shirt looking at a tablet. Overlaid on the image are various digital graphics: a Siemens logo in the top right, a '24/7' circular icon, a 'NEWS' section with a person icon, a 'Home' button, and a network diagram with three nodes. The text 'Industry Online Support' is also visible. The overall theme is industrial digitalization.

SIEMENS

LOGO! 8.3 – Modbus/TCP communication with a 7KN Powercenter 1000 as an example

LOGO!

<https://support.industry.siemens.com/cs/ww/en/view/109813923>

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1 Introduction

1.1 Overview

This user's example offers you a fully configured connection for LOGO! 8 (hereinafter: LOGO!) for communicating with a Modbus/TCP-capable device, a SENTRON 7KN Powercenter 1000 data transceiver (hereinafter: POC1000).

The integrated functions of LOGO! offer many additional possibilities to quickly and easily solve applications in various areas.

With LOGO!, prefabricated function blocks support project creation, e.g. a week timer, pulse generator, astro timer, seasonal time switch, stopwatch and simple logic gates.

The free LOGO! Web Editor (LWE) lets you create user-defined web pages for monitoring and control.

Advantages of LOGO! versus a conventional electrical installation

Using LOGO! offers you the following advantages:

- Extensibility of the software program to include further tasks. In addition to the basic task description, additional independent subtasks can be configured depending on the application. For this purpose, LOGO! can be expanded with a wide variety of expansion modules.
- Straightforward, star-shaped arrangement of the wiring of the components.
- Use of simple switches (circuit breakers) or pushbuttons.
- Fewer components are required compared to a conventional solution.
- Communication options via Modbus/TCP, S7 connection, and KNX.

Advantages of circuit protection devices with communication and metering functions

Use of a POC1000 with communication-capable circuit protection devices offers you the following benefits:

- Increased system availability thanks to early reaction to warning messages
- Easier troubleshooting with reporting of the original cause
- Increased transparency thanks to measurement of values such as current, voltage, active power, and many more.
- Better maintenance scheduling thanks to uptime counters and trigger counters

Circuit protection devices with communications and metering functions are an essential component of the SENTRON digitalization solutions.

Their compact form factor makes these circuit protection devices ideal for the retrofitting market, but also for new projects. The system can be commissioned easily with the SENTRON powerconfig PC software, the SENTRON powerconfig app for mobile devices, or with smartphones.

At the core of the system of communication-capable circuit protection devices is the 7KN Powercenter 1000 data transceiver. It collects measured values from the linked circuit protection devices and transmits them to higher-level systems. The measured values from up to 24 communication-enabled SENTRON devices are transmitted wirelessly to a 7KN Powercenter.

For more details on the components used, see [Links and literature](#) .

Target group

This application example is intended for specialists in electrical installations or automation.

1.2 Task description

The aim of this article is to demonstrate the essential principle of operation of the Modbus/TCP communication links between a LOGO! and a POC1000.

A LOGO! will communicate with a POC1000 via Modbus/TCP and receive various measured values, process them, and display them on a user-defined web page. These measured values are received by the LOGO! from the 5SL6 COM circuit breakers that are connected to the POC1000.

Successful initial commissioning of the POC1000 and the communication-enabled 5SL6 COM circuit breakers with the SENTRON powerconfig PC software (or the SENTRON powerconfig app) is a prerequisite for using this application example.

For more on this, see the following article and the device-specific operating manuals:

<https://support.industry.siemens.com/cs/ww/en/view/109805229>

1.3 Components used

This application example was created using the following hardware and software components:

Table 1-1

Component	Quantity	Item number	Note
LOGO! 230RCEo, logic module, PS/IO: 115V/230V/relay	1	6ED1052-2FB08-0BA1	230V power supply
7KN Powercenter 1000 data transceiver	1	7KN1110-0MC00	See manual (6)
Circuit breaker with metering function, communication AC 230V 6kA, 1+N pole, B, 16A	2	5SL6016-6MC	See manual (5)
LOGO! SOFT Comfort V8		6ED1058-0BA08-0YA1	-
LOGO! SOFT Comfort V8.3.1 Upgrade		6ED1058-0CA08-0YE1	see: https://support.industry.siemens.com/cs/ww/en/view/109811402
LOGO! Web Editor V1.1.1		-	
SETRON powerconfig app		-	See Apple App Store and Google Play Store

The listed components can be obtained from the [Siemens Industry Mall](#), for example.

This application example consists of the following components:

Table 1-2

Component	File name	Note
Documentation		This document
LOGO! switching program	7KN Powercenter 1000 & LOGO! 8.3 easy v02.snp	
LOGO! web pages	7KN Powercenter 1000 & LOGO! 8.3 easy v02.snp	

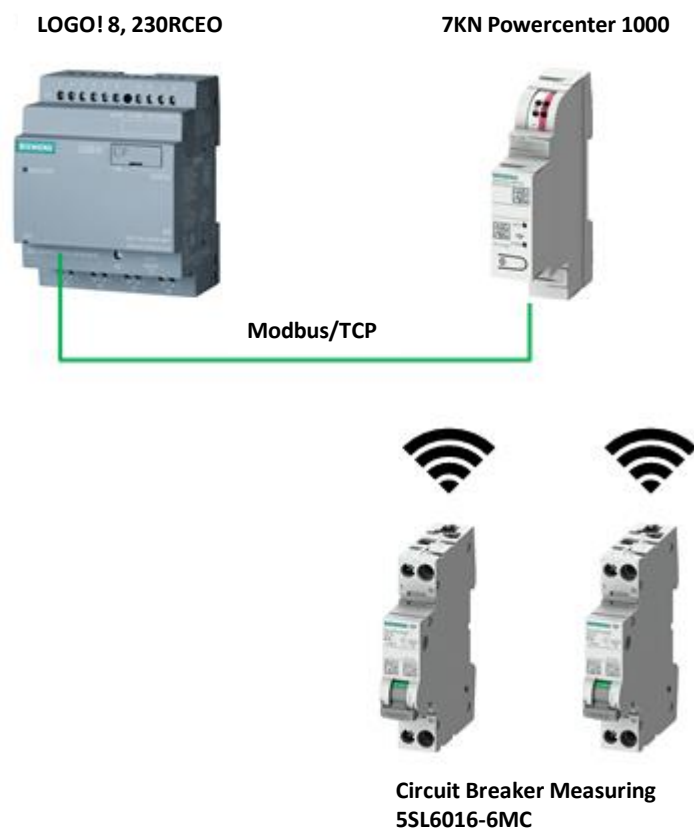
2 Engineering

2.1 Hardware configuration

The hardware configuration consists of a LOGO! in conjunction with a POC1000. The two are connected by a standard Ethernet cable.

Data exchange between the communication-enabled 5SL6 COM circuit breakers and the POC1000 takes place via a radio link. The POC1000 can communicate with up to 24 end devices. Coupling of the end devices with the POC1000 using the powerconfig mobile app must take place as an initial prerequisite once the devices are assembled and connected.

Figure 2-1



Note

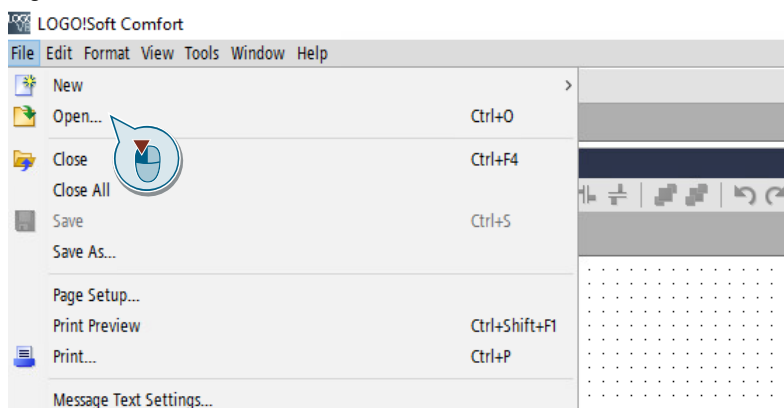
For further information on assembly and wiring, please refer to the manual ([3](#)), chapter: ([LOGO! installation and wiring](#)).

Follow the connection instructions in the product information supplied with your device.

2.2 Opening and adapting the application example

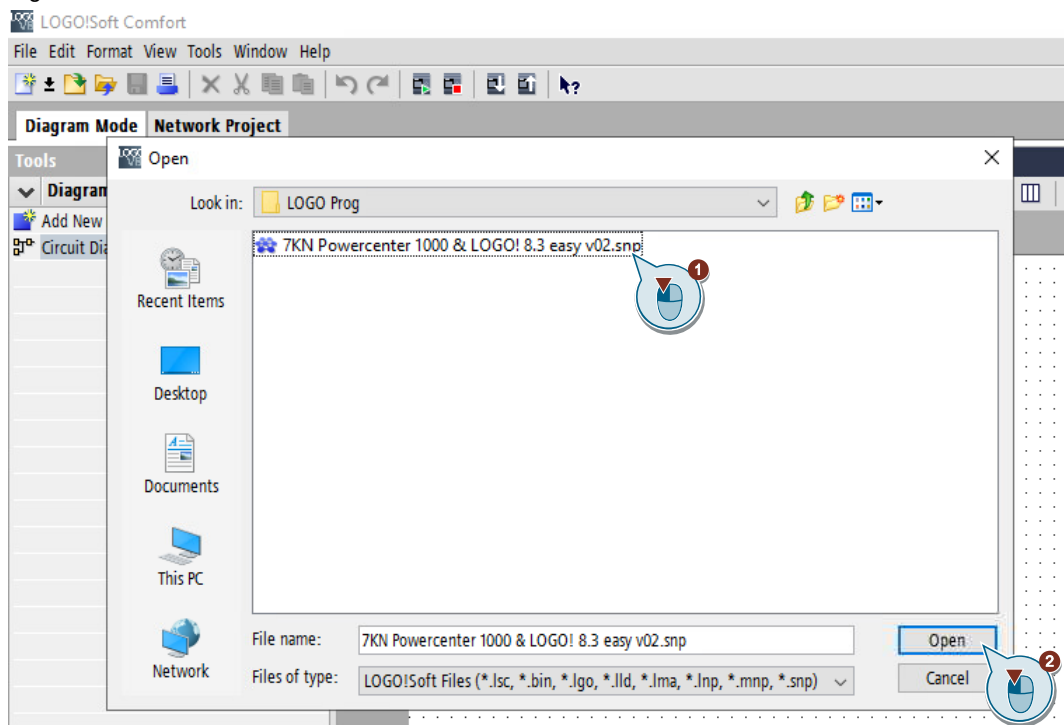
1. Extract the included ZIP file with the LOGO! switching program and the LWE program.
2. Launch LOGO!Soft Comfort V8.3.1
3. Click on "File" and on "Open".

Figure 2-2



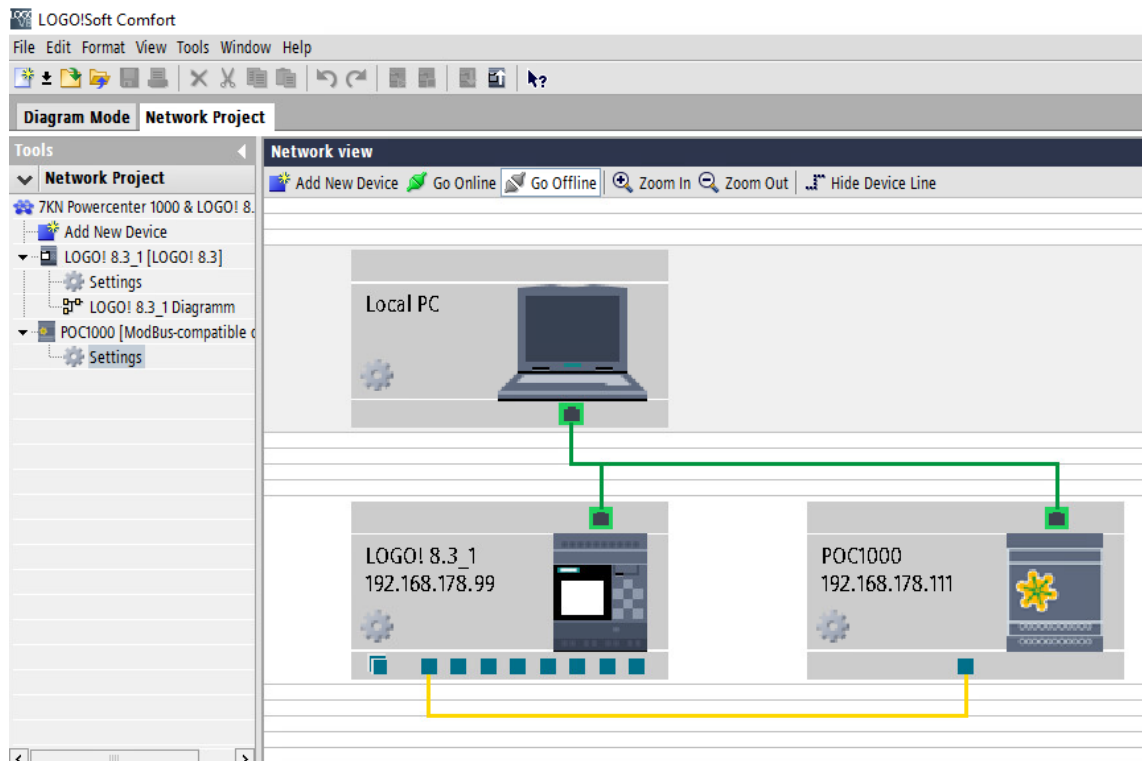
4. Select the project "47114711_7KN Powercenter 1000 & LOGO! 8.3 easy V02.snp".

Figure 2-3



5. After opening the project, switch to the network project view. Here you can see the network components involved: Local PC; LOGO! and POC1000.

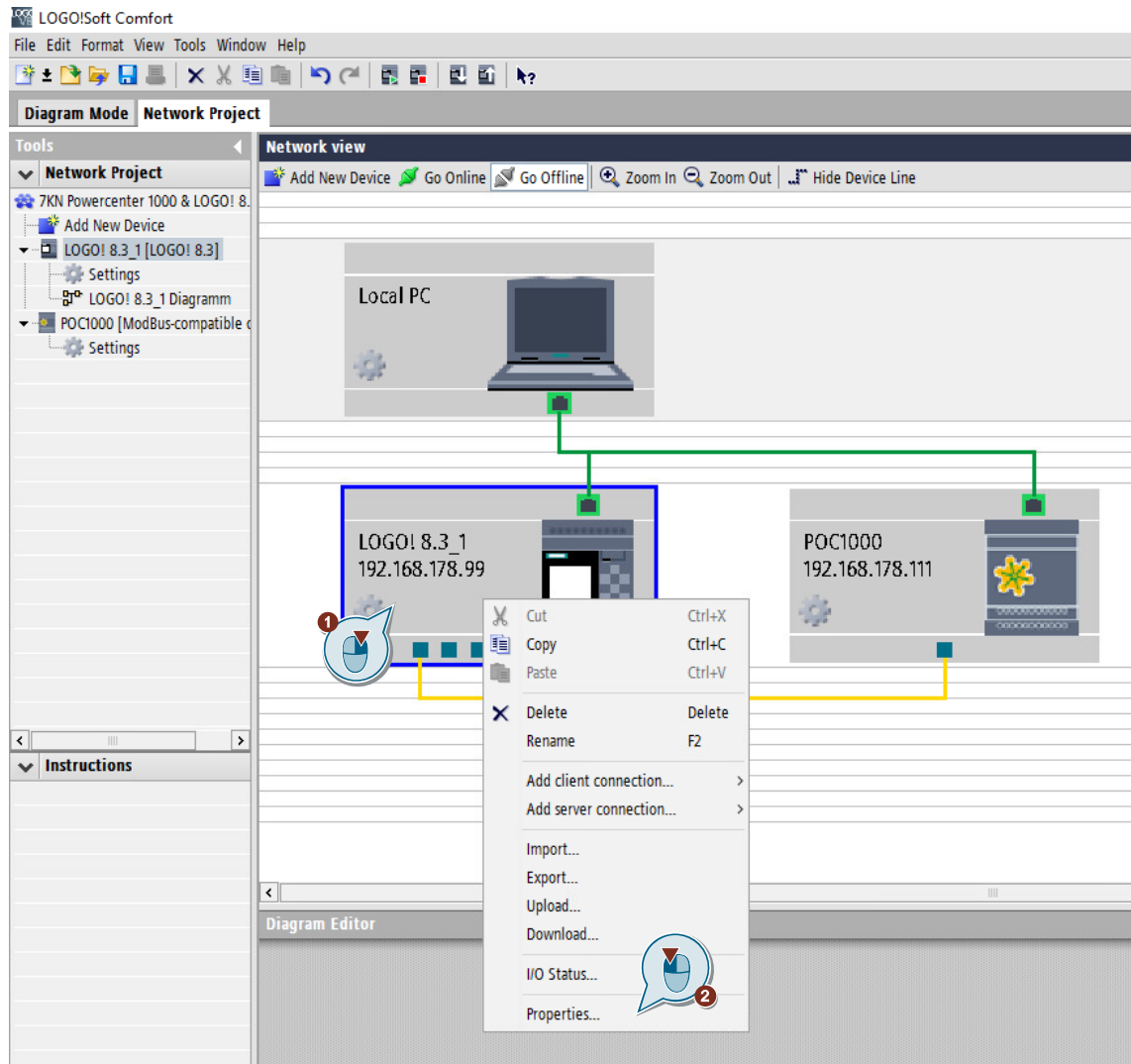
Figure 2-4



6. You can modify the IP addresses if necessary by selecting the LOGO! and/or the POC1000. To do this, select the component in question, switch to the Properties dialog, and change the IP addresses for both components.

- You can modify the IP settings of the LOGO! by right-clicking on the LOGO! and selecting Properties from the context menu.

Figure 2-5



8. Also make sure that S7 access and Modbus access are allowed.

Figure 2-6

The screenshot shows the 'LOGO! settings' window with the 'Offline settings' tab selected. The left sidebar lists various settings categories, with 'General' selected. The main area is divided into sections: 'Name settings', 'IP settings', and 'Ethernet connections'. The 'IP settings' section is highlighted with a red box and contains the following fields: IP Address (192.168.178. 99), Subnet Mask (255.255.255. 0), Default gateway (192.168.178. 1), and a checkbox for 'Customize DNS Server' which is unchecked. Below this, the 'Ethernet connections' section is also highlighted with a red box and contains two checked checkboxes: 'Allow S7 access' and 'Allow Modbus access'. To the right of these checkboxes are two yellow warning icons with exclamation marks. The first warning icon is next to the text 'Enabling the S7 access will open the insecure port 102.' and the second warning icon is next to the text 'Enabling the Modbus access will open the insecure port 502~510.' At the bottom right of the window are 'OK', 'Cancel', and 'Help' buttons.

9. Proceed in the same manner to modify the network settings on the POC 1000.

Figure 2-7

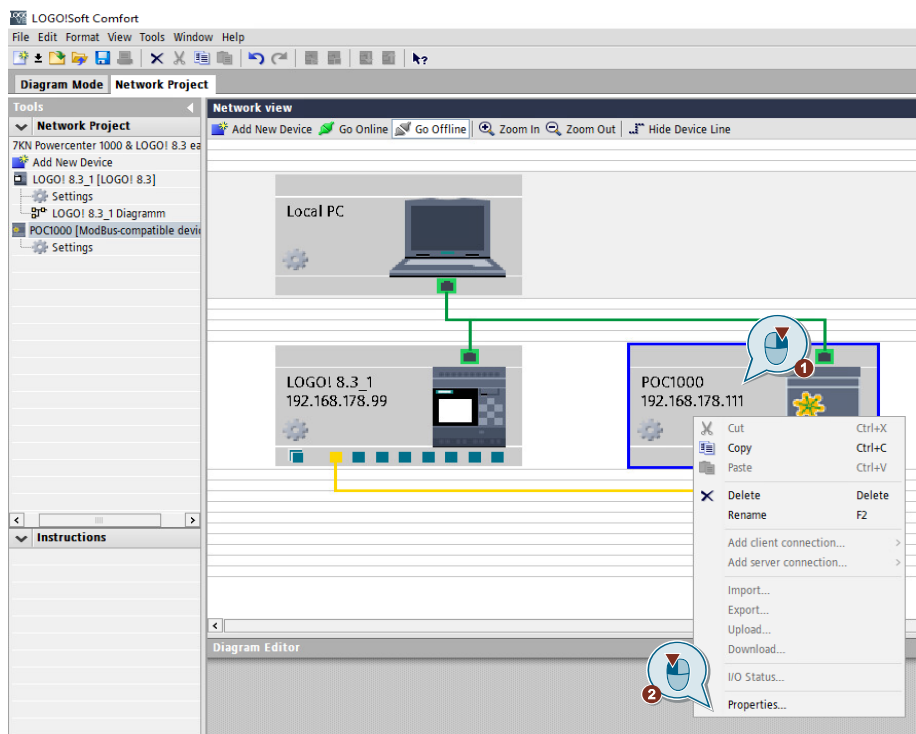
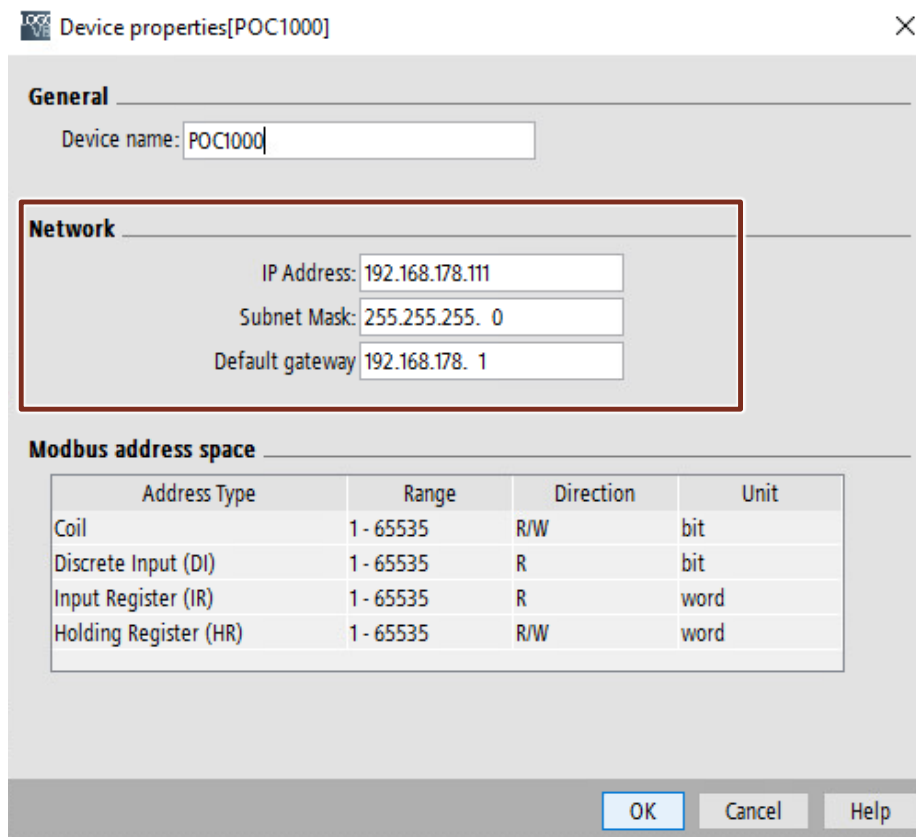


Figure 2-8

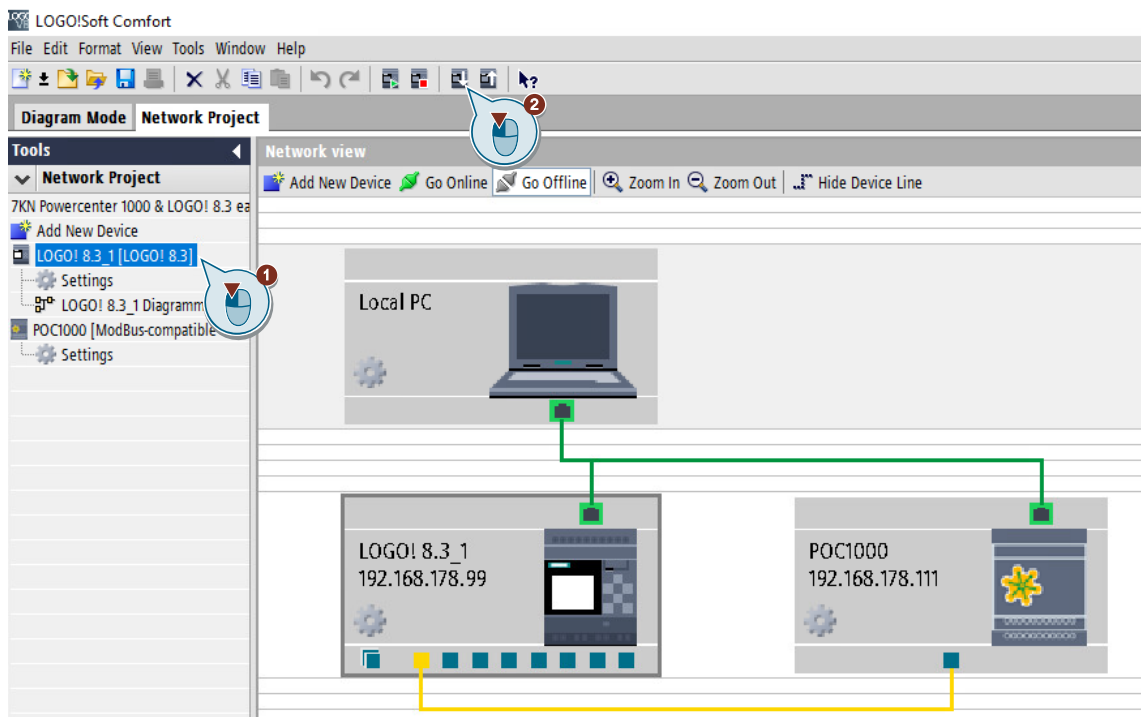


2.3 Loading application example with LOGO! SOFT Comfort V8.3.1

The LOGO! is connected, ready for operation and available on the network. To download the application to the LOGO! controller, proceed as follows:

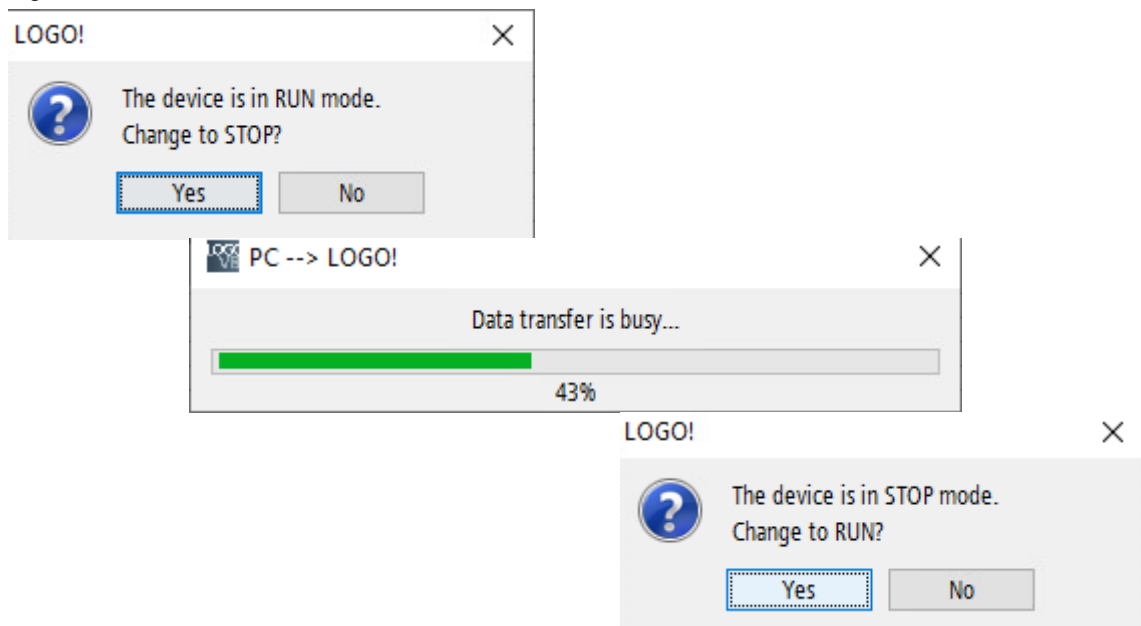
1. Highlight the LOGO! controller.
2. Click the Download button.

Figure 2-9



3. Confirm the dialog prompt about the RUN/STOP and STOP/RUN change.

Figure 2-10

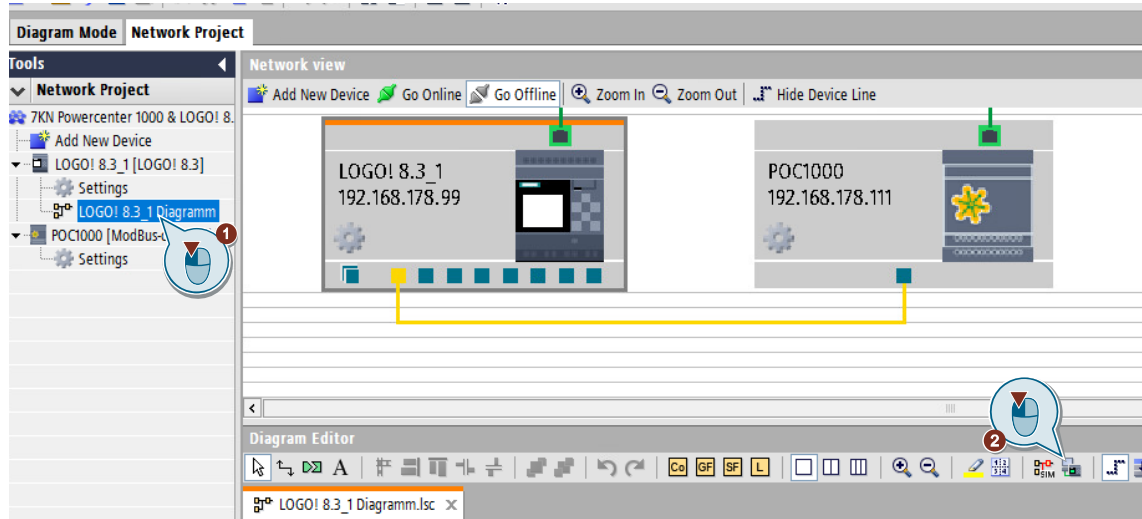


2.4 Testing the application example

Change to the chart page with the LOGIC blocks. This is where various measured values are read from the communication-enabled 5SL6 COM circuit breakers (named Unit 1 and Unit 2) and interconnected with flags.

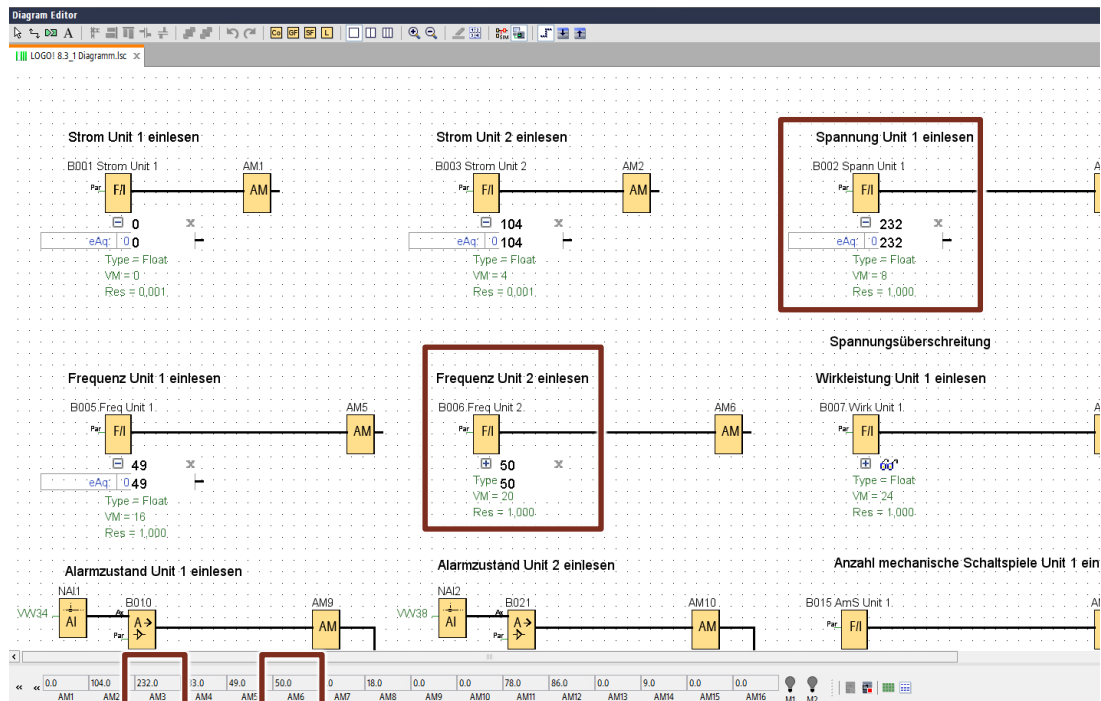
1. Change to Online Test to verify proper functioning.

Figure 2-11



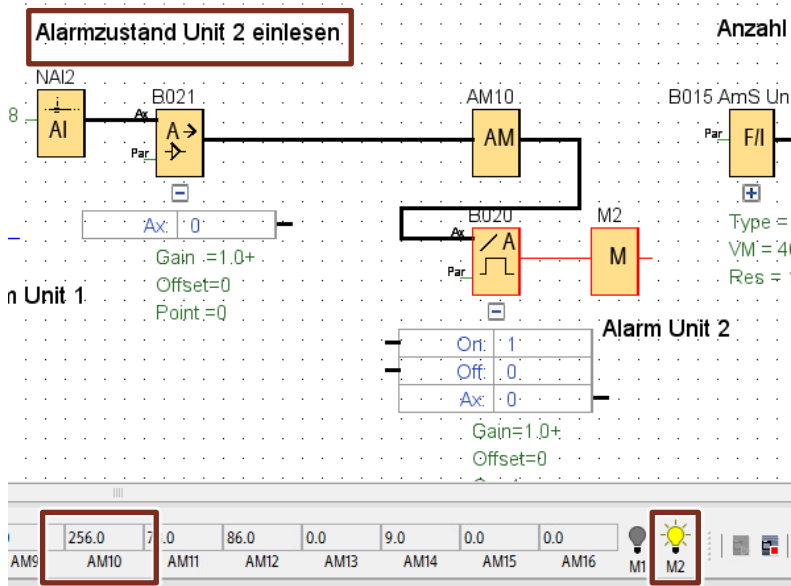
2. Assuming the application example has been implemented successfully, you can now see the voltage and frequency of the 5SL6 COM in the online view.

Figure 2-12



3. If an alarm status message from a 5SL6 COM is pending in the POC1000 (e.g. current limit exceeded), this alarm status for Unit 1 (first 5SL6 COM) can be seen with flag M1 and, for Unit 2 (second 5SL6 COM), with flag M2. An alarm value is pending if the cells AM9 (Unit 1) and/or AM10 (Unit 2) have a value not equal to 0.

Figure 2-13

**Note**

An alarm value can be tested, for example, via the SENTRON powerconfig app in the "Parameters" menu → "Alarm settings". To do this, go to "Alarm settings" in the app and set the alarm "Temperature alarm on/off" to "enabled" and the "Limit value temperature exceeded (degrees C)" to a value of 25, for example. A "Temperature alarm active" alarm will occur.

2.5 Loading the application example with LOGO! Web Editor V1.1.1

This application example also includes a web application that graphically displays selected measured values from the 5SL6 COM. To use the web application, LOGO! Web Editor V1.1.1 must be installed.

The LOGO! controller must also be equipped with a micro SD card. For more on this, see the instructions in the LOGO! system manual ([3](#)).

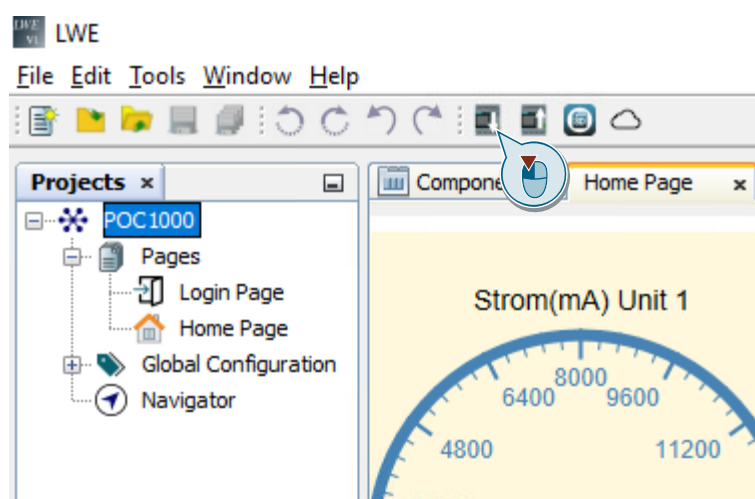
1. Open the project "POC1000LOGO8" with the LOGO! Web Editor V1.1.1.

Figure 2-14



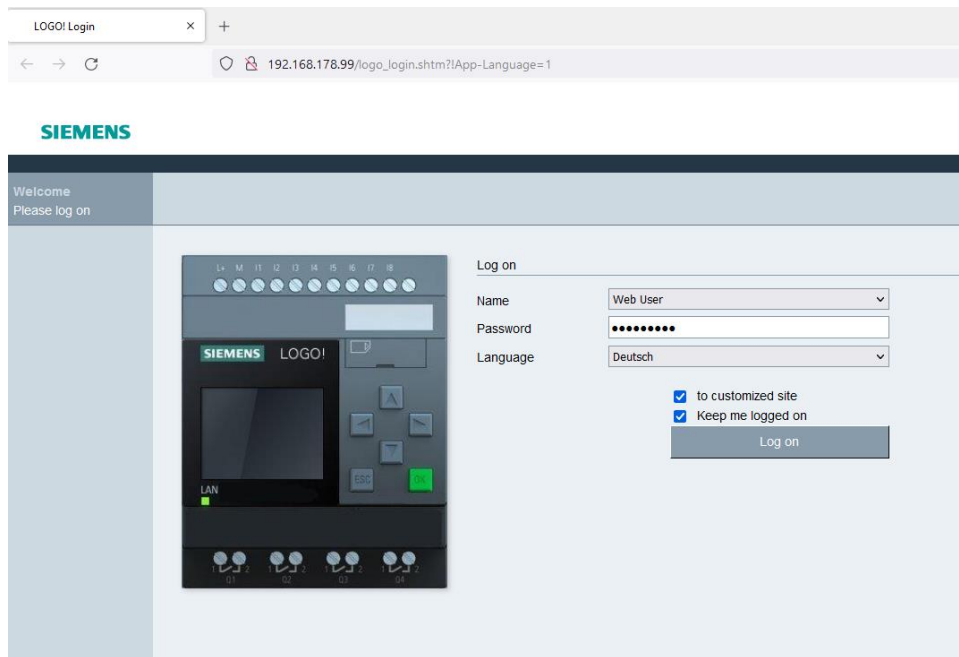
2. Download the project to the micro SD card of the LOGO!.

Figure 2-15



3. In a web browser, enter the IP address of the LOGO! to open its user-defined web page, then sign in. The default password is: **Logo**

Figure 2-16



4. You will see another selection of measured values such as current, voltage, frequency and alarm status in graphical form.

Figure 2-17



3 Useful information

3.1 Modbus TCP communication

A POC1000 supports up to 3 simultaneous Modbus TCP connections via the Ethernet port, functioning in the process as a Modbus TCP server. A Modbus TCP server provides a Modbus TCP client (here: LOGO!) with data. The addressing of information from the individual measurement- and communication-enabled SENTRON circuit protection devices via Modbus TCP occurs on 2 levels:

1. Via the device address of the individual devices (Unit 1 = 1; Unit 2 = 2).
2. Via the registry address within the devices.

The measurement variable or a status is addressed with the registry address. Besides the registry address, there is also other descriptive information that is specified, such as the data format (fixed-point number, floating-point number, etc).

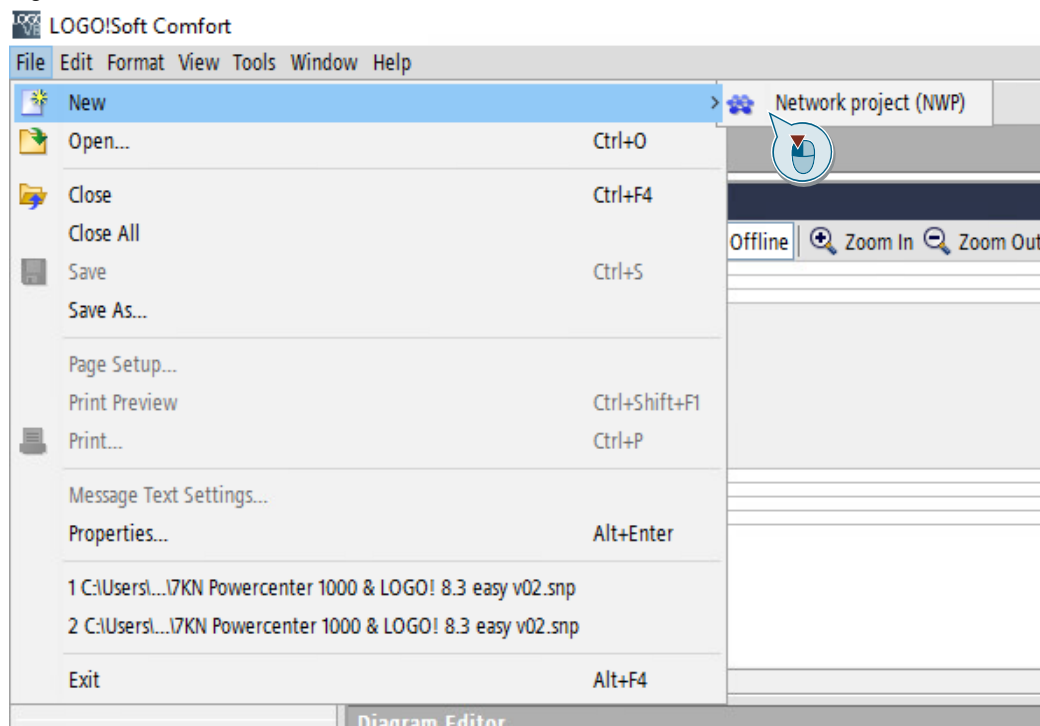
For more on this, see also the information from the system manual for circuit protection devices with communication and metering functions:

<https://support.industry.siemens.com/cs/ww/en/view/109791806>

3.2 Creating a LOGO! → POC1000 Modbus TCP connection

1. Use the LOGO! SOFT Comfort V8.3 software to create a new network project.

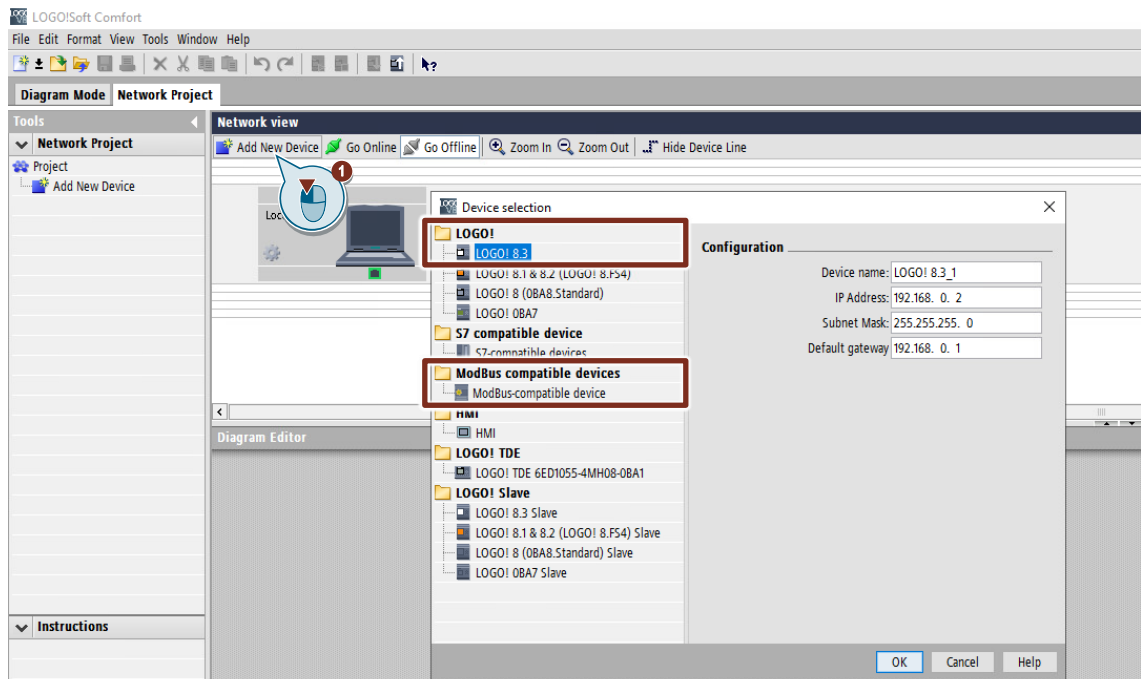
Figure 3-1



3 Useful information

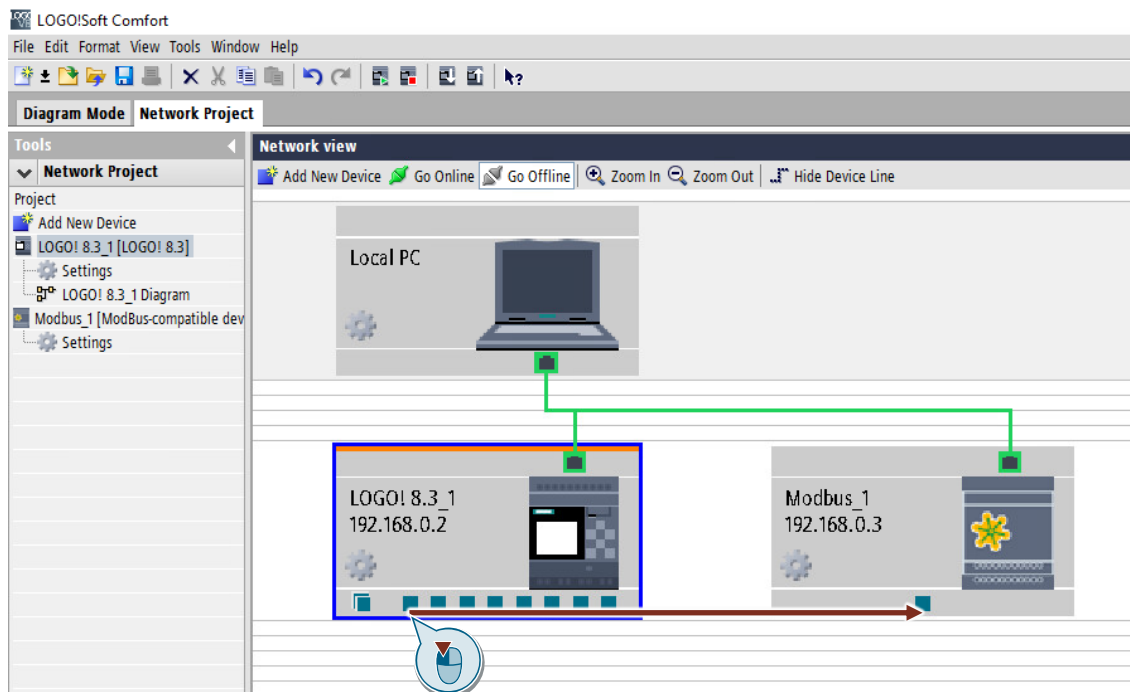
2. Add a LOGO! controller and a Modbus-compatible device in succession.

Figure 3-2



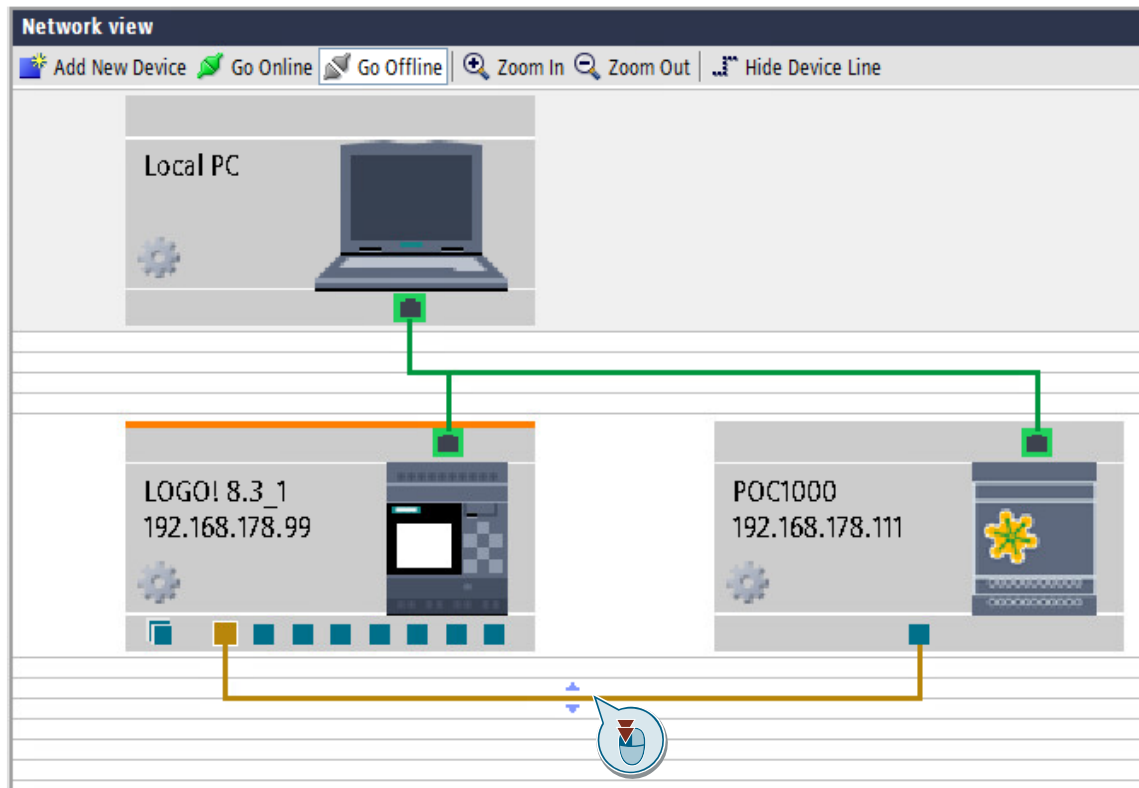
3. Draw a communication line from left (LOGO!) to right (POC1000).

Figure 3-3



4. Check the correct communication definition (LOGO! is a Modbus TCP client and the POC1000 is the Modbus TCP server) by double-clicking on the communication line.

Figure 3-4



5. The Modbus TCP connection window opens.

Figure 3-5

Modbus Connection

Device name: LOGO! 8.3_1
 IP Address: 192.168. 0. 2
 PORT: Assigned
☐ Client ☐ Server

Device name: Modbus_1
 IP Address: 192.168. 0. 3
 PORT: 502
☐ Client ☒ Server

ID	Start Address	Length	Direction	Start Address	Length	Unit ID
1	V 0.0		Coil			255

Synchronize interval

☒ Enable customized interval

0 Hours 0 Minutes 0 Seconds 80 Milliseconds

OK Cancel Help

The device address and the registry addresses of the relevant measurement variables are entered in the Modbus connection window. The port address of the Modbus TCP server is 502 by default.

The synchronization interval is also specified in this window.

Note

For measurement variables and status information, a minute interval is typically more than enough for a refresh frequency.

The LOGO! has a variable memory, VM. By indicating the proper memory address (starting with VW0), the Modbus TCP server stores data via the device address (Unit ID) and the registry address in the variable memory of the LOGO!.

Example:

Using device address 1 (Unit ID = 1) and the registry address 3076 for the first 5SL6 COM, the instantaneous current value from the 5SL6 COM is saved starting at memory address VW0. Since registry address 3076 holds a total of 2 words, the next memory address can begin at VW4.

Using device address 2 (Unit ID = 2) and registry address 3084, the mains frequency for the second 5SL6 COM is saved to variable memory VW20 of the LOGO!.

Figure 3-6

Modbus Connection

Device name: LOGO! 8.3_1

IP Address: 192.168.178.99

PORT: Assigned

☐ Client ☐ Server

Device name: POC1000

IP Address: 192.168.178.111

PORT: 502

☐ Client ☐ Server

Data transfer

ID	Start Address	Length	Direction	Start Address	Length	Unit ID
1	VW ▼ 0	2 words	<--	HR ▼ 3076	2 words	1
2	VW ▼ 4	2 words	<--	HR ▼ 3076	2 words	2
3	VW ▼ 8	2 words	<--	HR ▼ 3082	2 words	1
4	VW ▼ 12	2 words	<--	HR ▼ 3082	2 words	2
5	VW ▼ 16	2 words	<--	HR ▼ 3084	2 words	1
6	VW ▼ 20	2 words	<--	HR ▼ 3084	2 words	2
7	VW ▼ 24	2 words	<--	HR ▼ 3086	2 words	1
8	VW ▼ 28	2 words	<--	HR ▼ 3086	2 words	2
9	VW ▼ 32	2 words	<--	HR ▼ 2560	2 words	1
10	VW ▼ 36	2 words	<--	HR ▼ 2560	2 words	2
11	VW ▼ 40	2 words	<--	HR ▼ 2593	2 words	1
12	VW ▼ 44	2 words	<--	HR ▼ 2593	2 words	2
13	VW ▼ 48	2 words	<--	HR ▼ 2602	2 words	1
14	VW ▼ 52	2 words	<--	HR ▼ 2602	2 words	2
15	VW ▼ 56	2 words	<--	HR ▼ 2624	2 words	1
16	VW ▼ 60	2 words	<--	HR ▼ 2624	2 words	2

Synchronize interval

☒ Enable customized interval

Hours
 Minutes
 Seconds
 Milliseconds

OK Cancel Help

3.3 Processing measured variables in the LOGO!

Measured variables that are read via the variable memory of the LOGO! are processed in the chart editor. In the chart editor, it is possible to store logic ops in the LOGO! by combining a variety of function blocks.

To process the ranges for the respective values (configured previously in the Modbus connection table), you have the choice of using an "Analog network input" or a "Floating-point/integer converter" function block.

The advantage of the "Floating-point/integer converter" is the ability to directly observe the received numerical values, as well as the ability to adjust the decimal places.

Example:

The frequency of the first 5SL6 COM is stored in the variable memory starting at VW16. Double-clicking on the "Floating-point/integer converter" (added to the chart in a previous step) takes the user to the converter's parameter properties (Figure 3-8). The desired variable memory (here: VW16) and the resolution can be entered there. The resolution of "1" yields the expected frequency of 50 Hz.

Figure 3-7

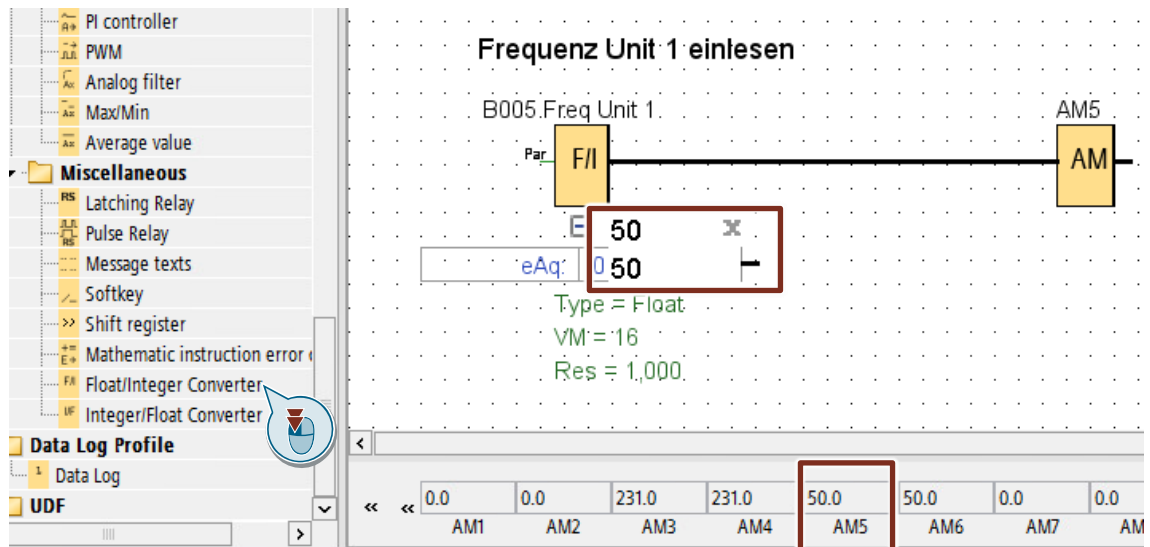
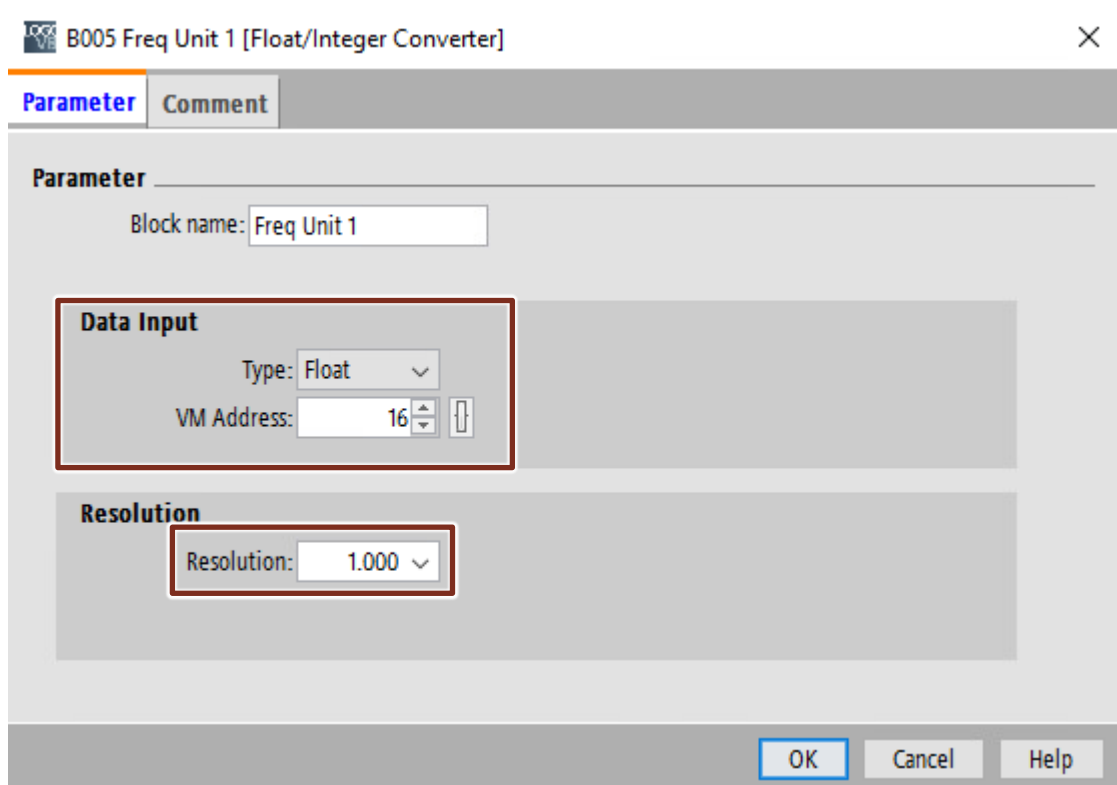
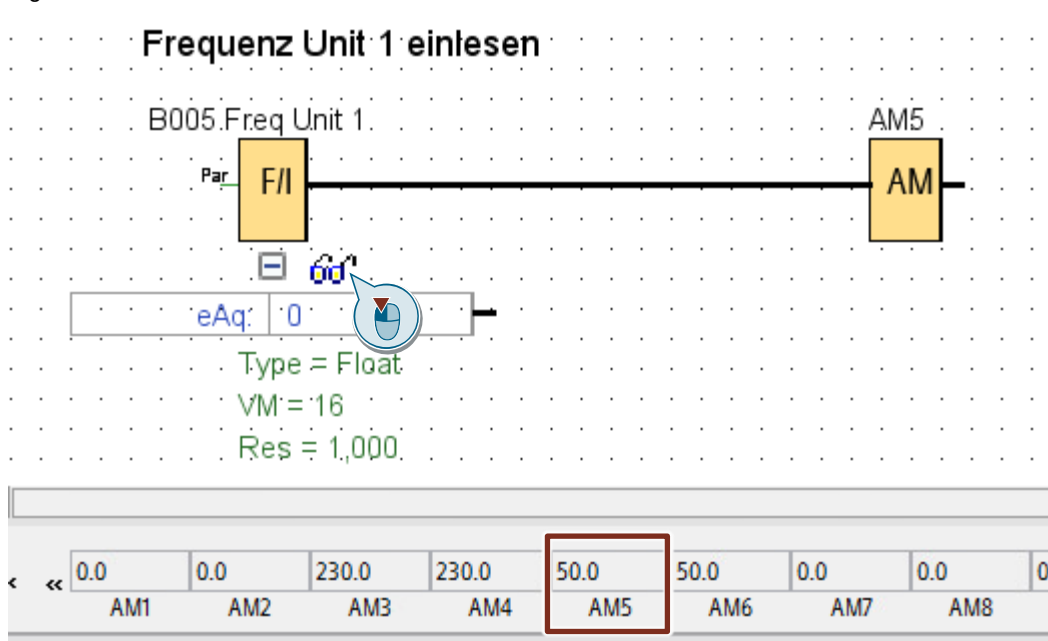


Figure 3-8



In the "Online Test" status of the LOGO!, the eyeglasses icon can be used to observe the converted counter value directly.

Figure 3-9



The logic op with an analog flag adds a status line to the view.

4 Appendix

4.1 Service and support

Industry Online Support

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4.3 Links and literature

Table 4-1

No.	Topic
\1\	Siemens Industry Online Support https://support.industry.siemens.com
\2\	Link to the article page of the application example https://support.industry.siemens.com/cs/ww/en/view/109813923
\3\	LOGO! system manual https://support.industry.siemens.com/cs/ww/en/view/109741041
\4\	LOGO! 8.3 and 7KN Powercenter 1000 – Communication in Modbus TCP IP https://support.industry.siemens.com/cs/ww/en/view/109805229
\5\	System Manual - Circuit protection devices with communication and metering function https://support.industry.siemens.com/cs/ww/en/view/109791806
\6\	7KN POWERCENTER 1000 https://support.industry.siemens.com/cs/ww/en/view/109793297
\7\	Support of Ethernet/TCP communication connections https://support.industry.siemens.com/cs/ww/en/view/61630471

4.4 Change documentation

Table 4-2

Version	Date	Change
V1.0	10/2022	First version